Variables & Scripts in Postman

Variables in Postman

What are Variables?

Variables are placeholders for values used in requests, scripts, or collections.

Why to use Variables?

- To avoid hardcoding values.
- To make scripts reusable and dynamic.
- To manage configurations for different environments.

Where Can Variables Be Used?

- In URLs, headers, request bodies, or scripts.

Types of variables in Postman:

1. **Global Variable** (Accessible everywhere):
   url_global = https://reqres.in
2. **Environment Variable** (Specific to an environment like QA or Prod):
   url_qa_env = https://reqres.in
3. **Collection Variable** (Specific to a collection of requests):
   url_collect = https://reqres.in
4. **Local Variable** (Specific to a single request):
   url_local = https://reqres.in
5. **Data Variable** (Values provided through CSV/JSON files in the collection runner).

Accessing Variables in URLs:
Use **{{variable_name}}** to reference variables in the request URL.

Example:

{{url_global}}/api/users?page=2

{{url_qa_env}}/api/users?page=2

{{url_collect}}/api/users?page=2

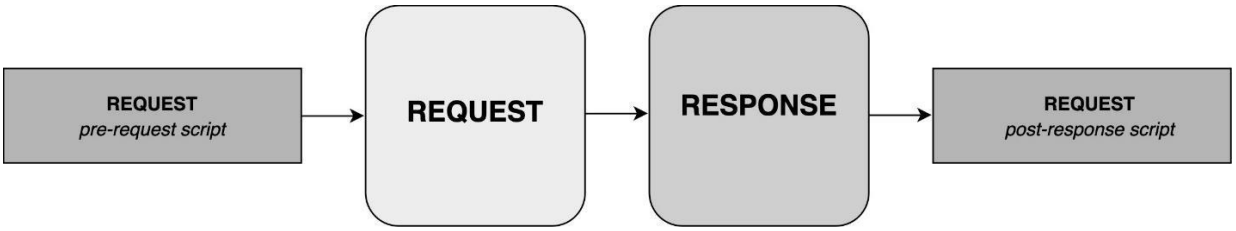
{{url_local}}/api/users?page=2

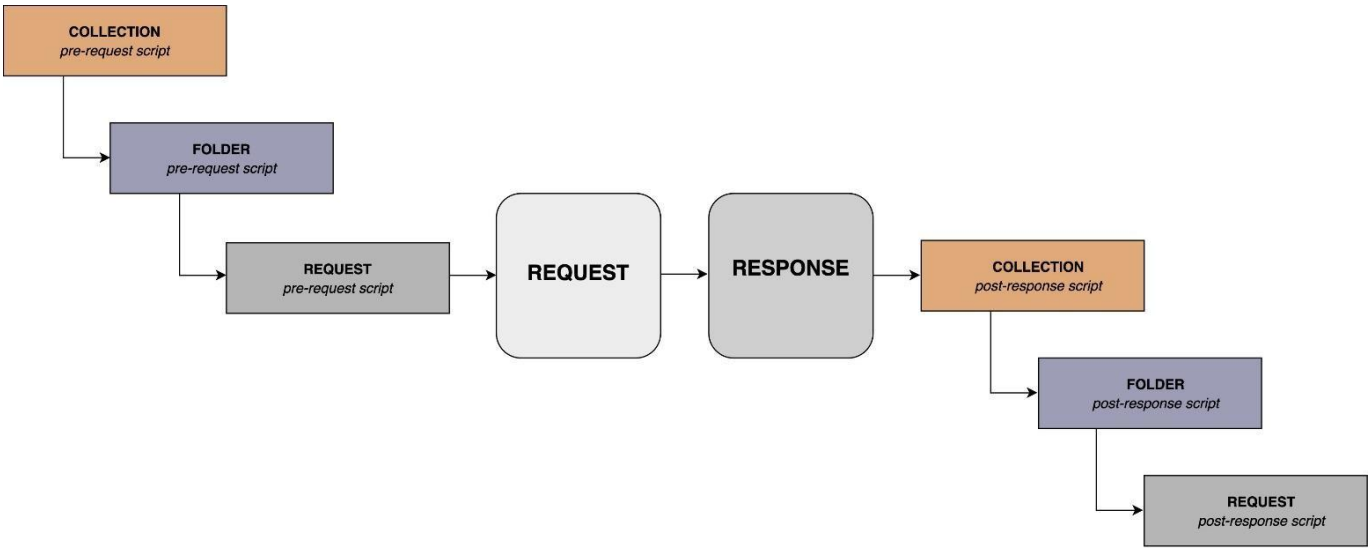
Postman Scripts

Postman scripts can be written at different stages:

1. **Pre-request scripts:** Run before the request is sent.
2. **Post-response scripts:** Run after receiving the response.

Workflow: Pre-request Script → Request → Response → Post-response





- **Collection:** A group of requests.
- **Folder:** Subgroups within a collection.
- **Request:** Individual HTTP requests (GET, POST, etc.).

Creating and Managing Variables with Scripts

Creating Variables in Pre-request Scripts

You can set variables before sending a request using JavaScript.

- **Global Variable:**

```
pm.globals.set("userid_global", "2");
```

- **Environment Variable:**

```
pm.environment.set("userid_qa_env", "2");
```

- **Collection Variable:**

```
pm.collectionVariables.set("userid_collect", "2");
```

- **Local Variable:**

```
pm.variables.set("userid_local", "2");
```

Retrieving and Removing Variables in Post-response Scripts

- **Retrieve a Variable Value:**

```
console.log(pm.globals.get("userid_global"));
console.log(pm.environment.get("userid_qa_env"));
console.log(pm.collectionVariables.get("userid_collect"));
console.log(pm.variables.get("userid_local"));
```

- **Unset (Remove) a Variable:**

```
pm.globals.unset("userid_global");
pm.environment.unset("userid_qa_env");
pm.collectionVariables.unset("userid_collect");
pm.variables.unset("userid_local");
```

Using Variables Inside the Request Body

You can set and use variables in the request body to make dynamic requests.

1. **Create Local Variables in Pre-request Script:**

```
pm.variables.set("name", "Pavan");
pm.variables.set("job", "Trainer");
```

2. **POST Request Example:**

URL: https://reqres.in/api/users

Request Body:

```
{
   "name": "{{name}}",
   "job": "{{job}}"
}
```

Here, **{{name}}** and **{{job}}** will be replaced with the values "Pavan" and "Trainer" dynamically.